A Multi-Modal Medical Image Fusion Technique Using Structural Similarity Index Measure and Correlation Coefficient

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***Abstract* — This research paper introduces an innovative approach (M3IF-SSIM-CC) to multimodal medical image fusion (M3IF) using the Non-Subsampled Contourlet Transform (NSCT), specifically targeting enhanced clinical decision-making through improved diagnostic imagery. M3IF-SSIM-CC integrates crucial diagnostic information from Computed Tomography (CT) and Magnetic Resonance Imaging (MRI) scans. Initially, NSCT decomposition is applied independently to CT and MRI images, generating low-pass (LP) and high-pass (HP) sub-bands. The LP sub-bands undergo fusion through a correlation-based rule designed to maintain structural coherence and maximum information retention. Concurrently, the HP sub-bands are combined using a similarity-based fusion technique aimed at preserving critical edges and details. Following the fusion of these sub-bands, inverse NSCT reconstructs a comprehensive image that encapsulates anatomical and pathological features from both modalities. Rigorous evaluation, conducted using various quantitative metrics and clinical assessments, demonstrates the superiority of the M3IF-SSIM-CC over traditional fusion techniques. Results show significant improvements in image clarity, preservation of detail, and overall diagnostic accuracy, facilitating more precise and confident clinical interpretations. M3IF-SSIM-CC represents a meaningful advancement in medical imaging technologies, providing clinicians with enhanced diagnostic tools for improved patient outcomes.**

Keywords—image fusion, multi-modal medical image fusion, SSIM, CC.

I. INTRODUCTION

Image fusion aims to create a new image by integrating the characteristics of many input images. The whole output, including all essential data inside a one image, is more accurate and informative than any individual source image. The dual aims of image fusion, namely reducing data volume and improving image quality for both human and machine observers, are well-recognized. One application of image fusion is multi-sensor image fusion, which involves consolidating pertinent data from several source images into a single output image. Image fusion, an intriguing field within image processing, utilizes many approaches to produce an integrated image for analysis and diagnosis. Image fusion requires the execution of five fundamental steps: registration, decomposition, fusion rules, reconstruction, and assessment methods [1].

Image fusion encompasses several categories based on the dataset, including multi-focus, multi-sensor, multi-scale, multi-spectral, and multi-temporal. A single input image that retains all relevant information may be produced by integrating several input images with varying focal points using multi-focus image fusion methods. Medical diagnostic applications use multi-modal approaches for image fusion, integrating data from several sources to provide a more comprehensive and coherent image [2]. This data may be invisible to the naked eye. It is possible to locate anomalies using data derived from composite pictures precisely. The multi-scale image fusion approach efficiently combines Low Dynamic Range images with varying exposures into a single, more informative LDR image. Multi-spectral image fusion approaches use data integration in the spectral domain to leverage associated information and spatiotemporal correlations, yielding a coherent image. Information failure is mitigated, and multi-temporal image fusion methods acquire full data on all clinical parameters [3]. Fig 1 illustrates an example of multi-modal medical image fusion.

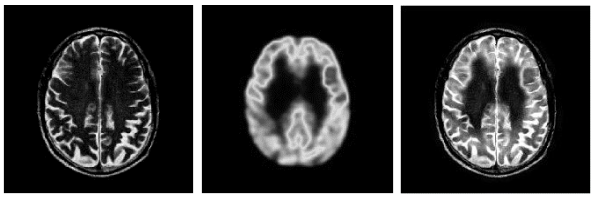


Fig 1. Example of multi-modal medical image fusion

Physicians rely on multimodal medical fused images for a variety of medical diseases; these images are vital for aiding in diagnosis, directing therapy, and detecting improvement. Medical professionals use many imaging modalities for illness diagnosis, including positron emission tomography (PET), T1-weighted and T2-weighted magnetic resonance imaging (MR-T1 and MR-T2), CT, magnetic resonance angiography (MRA), and MRI. These imaging methods may provide extensive data on a patient's health, are economically efficient, and user-friendly. While several benefits exist, defining them is a considerable challenge. Even proficient radiologists may have difficulties in identifying minute nodules or distinguishing between similar cancers. Manual illness screening requires significant time and effort. Image fusion has emerged as a very promising area of image processing, owing to its significant relevance in several applications, such as medical diagnosis and picture enhancement. Multimodal Medical Image Fusion improves medical image quality by amalgamating images acquired from several modalities. The result is a composite picture that exceeds the sum of its parts [4].

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This study aims to create a multimodal medical imaging approach that combines NSCT with SSIM and CC, using MRI and CT images as input. This hybrid technique employs a smart fusion methodology to optimize the information content in the fused image more effectively. It will then be compared with previous fusion procedures. The integration of NSCT with SSIM and CC via a smart fusion threshold-based approach improves the results of this hybrid technique by implementing a noise reduction strategy.

After the introduction, Section 2 of the article details all the relevant theories that informed this study. The technique is described in Section 3, along with an algorithm and a flow diagram. The article is concluded in Section 5, after which Section 4 delves into the outcomes of fusion.

II. Related Theory

A. Image Fusion using NSCT domain

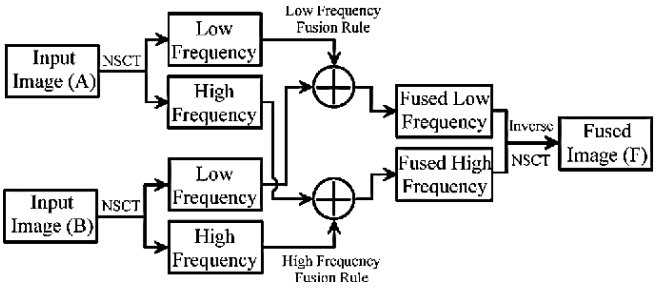


Fig 2. General image fusion modal using NSCT domain

The general image fusion modal using NSCT domain is shown in above Fig 2. This figure explains how the NSCT-based image fusion works. A and B are two input images that are processed using the NSCT. For images A and B, respectively, the NSCT decomposes each input image into low-frequency and high-frequency sub bands. To create a fused low-frequency sub band, the low-frequency sub bands (L and L') are joined using a chosen fusion rule during the image fusion process. This rule usually prioritizes maintaining the most significant attributes, such as the overall brightness or contrast. The high frequency sub bands (H and H') are similarly fused according to a different fusion rule, the goal of which is often to maintain fine texture and edge features. To create the fused NSCT representation, the low-frequency and high-frequency sub bands are fused together. The next step is to use the inverse NSCT (INSCT) on the combined low- and high-frequency sub bands to create a fused image. This will keep the best parts of both input images. Some of the latest research in the field of multi-modal medical image fusion using NSCT domain are discussed below:

To address the limitations of conventional medical image fusion techniques, including inadequate detail preservation, edge information loss, image distortion, and the substantial requirement for training data in deep learning, a novel multi-modal medical image fusion method is proposed [4]. This method is based on the VGG19 model and the NSCT, with the primary aim of leveraging the strengths of both the NSCT and the VGG19 model concurrently. A novel hybrid fusion scheme is introduced utilizing the NSST in conjunction with the stationary wavelet transform (SWT). During the initial phase, multimodal input images undergo thorough degradation through the application of NSST. The regional energy serves as an activity parameter for the fusion of high-frequency sub-band coefficients in NSST. The approximation sub-band of the NSST is fused utilizing the SWT method [5].

In the NSST framework, a new shrinkage function is presented in [5]. The SURE-LET idea is a powerful tool for using the multivariate shrinkage model. Both actual and synthetic CT images with noise are used to evaluate the outcomes. In most cases, the findings show that the recommended strategy produces better results [5]. Medical image fusion using the SWT is introduced in a novel multi-modality fusion method in [6]. The proposed method in [6] starts with breaking down source images into SWT-based approximation and detail layers. Afterwards, they use a local contrast fusion approach and Fuzzy Local Information C-means Clustering to reduce blurring, increase sensitivity, and keep quality assessment in each layer intact.

B. Structural Similarity Index Measure (SSIM)

SSIM measures the optical deterioration of an image due to things like data compression and transmission losses. It uses brightness, contrast, and structural similarity as its three main metrics to determine how similar two images are to one another. The value of SSIM ranges from 0 to 1. The whole visual quality is heavily dependent on each of these factors. To control the image fusion process in this manner, SSIM is used inside the fusion rule. After calculating the SSIM values of the input images, a weight is applied to each one according to a pre-established threshold. To make sure that the fused image has its high level of brightness, contrast, and structure, it is necessary to save the most relevant aspects from both images while minimizing perceived quality loss [7].

$$SSIM\ (HP, HP') = \frac{((2\mu_{HP}\mu_{HP'} + C_1) \times (2\sigma_{HPHP'} + C_2))}{((\mu_{HP}{}^2 + \mu_{HP'}{}^2 + C_1) \times (\sigma_{HP}{}^2 + \sigma_{HP'}{}^2 + C_2))} \quad (1)$$

μ_{HP} is the mean over a window in high pass sub band HP

$\mu_{HP'}$ is the mean over a window in high pass sub band HP

σ_{HP} is standard deviation over a window in high pass sub band HP

$\sigma_{HP'}$ is standard deviation over a window in high pass sub band HP'

$\sigma_{HPHP'}$ is co-variance over a window between high pass sub band HP and HP'.

C_1, and C_2 are constants.

C. Correlation Coefficient (CC)

In statistics, the CC is a useful metric for drawing connections between experimental results and predictions. To see how closely the actual numbers match the predictions, one may look at the computed value of the correlation coefficient. a positive correlation is indicated by a CC value of 1. Values around 0 show a poor connection or no linear relationship at all [8]. The threshold value T is set at 0.6 in this study.

$$\rho_{LPLP'} = \frac{Cov(LP, LP')}{\sigma_{HP}\sigma_{HP}{}'} \quad (2)$$

$\rho_{LPLP'}$ is the Pearson product-moment correlation coefficient.

$Cov(LP, LP')$ covariance of LP and LP'

σ_{HP} is the standard deviation of HP

$\sigma_{HP'}$ is the standard deviation of HP'

III. Proposed Methodology: M3IF-SSIM-CC

A. Proposed Algorithm

Step 1. Apply NSCT on input image 1 (CT image), A

Step 2. Apply NSCT on input image 2 (MRI image), B

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Step 3. CT image is decomposed into low pass sub bands (LP), and high pass sub bands (HP).

Step 4. MRI image is decomposed into low pass sub bands (LP'), and high pass sub bands (HP').

Step 5. Fusion of low pass sub bands:

By using a local correlation-based technique, the low pass sub bands (LP, LP') that have been acquired are fused using below formula. In this approach, utilizing a 4 × 4 block size, the low pass sub bands are used to calculate block wise correlation coefficients. The correlation coefficients are calculated using Eq 2. The median values of the low pass sub bands are denoted by LP and LP', respectively. The correlation-based fusion method compares the resultant correlation value (C) to a predetermined threshold value (T). Here is a correlation-based fusion strategy:

$$LP^{new} = \begin{cases} \max(LP, LP') & if(C \leq 0.6) \\ avg(LP, LP') & otherwise \end{cases}$$

where, max and avg stand for maximum and average values, respectively, and LP^{new} are the fused coefficients.

Step 6. Fusion of high pass sub bands:

By using a similarity-based technique (for analyzing brightness, contrast, and structure), the high pass sub bands (HP, HP') that have been acquired are fused using below formula. In this approach, utilizing a 4 × 4 block size, the high pass sub bands are used to calculate SSIM block wise. The SSIM are calculated using Eq 1. The median values of the high pass sub bands are denoted by HP and HP', respectively. The similarity-based fusion method compares the resultant SSIM value (SSIM) to a predetermined threshold value (T). Here is a similarity-based fusion strategy:

$$HP^{new} = \begin{cases} \max(HP, HP') & if(SSIM \leq 0.6) \\ avg(HP, HP') & otherwise \end{cases}$$

where, max and avg stand for maximum and average values, respectively, and HP^{new} are the fused coefficients.

Step 7. Inverse NSCT is applied to obtain the fused image as shown in below equation. G is the fused image.

$$\text{L} = \text{IDWT}\ (LP^{new}, HP^{new})$$

Step 8. L is the final fused image.

B. Significance of M3IF-SSIM-CC

The M3IF-SSIM-CC achieves improved outcomes through the application of SSIM and the CC with the incorporation of NSCT as shown in Fig 3. This hybrid integration of various metrics with NSCT improves the fusion results in terms of many factors like structural information, brightness, contrast, etc. The fusion rule uses a threshold that decides fusion rule of SSIM and CC. This fusion rule is capable of being applied in various image enhancement and restoration models, owing to the effectiveness of the method noise concept.

C. Proposed Flowchart: M3IF-SSIM-CC

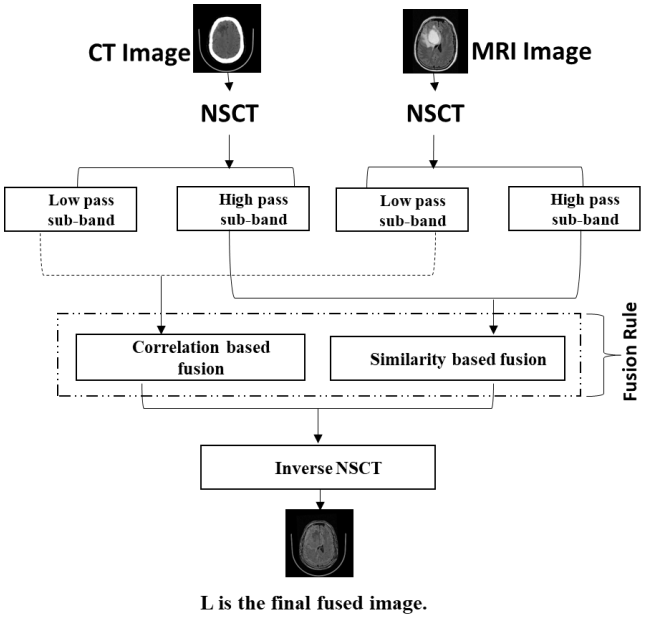
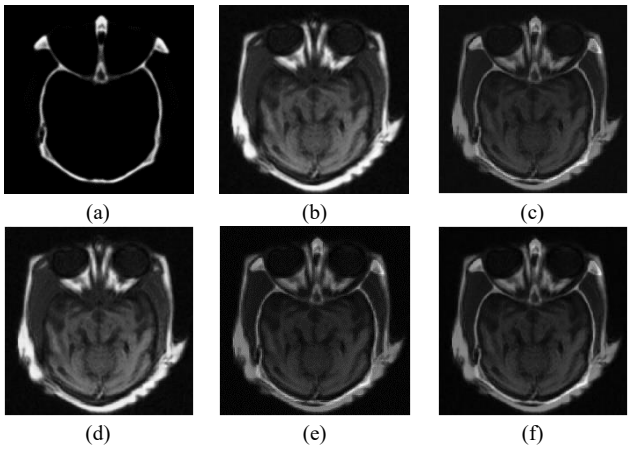


Fig 3. Flowchart of proposed methodology

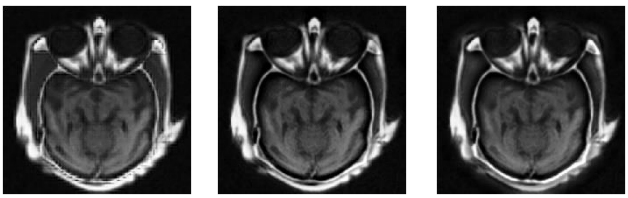
IV. Results And Discussion

Figures 4 (a, b) illustrate the datasets of CT and MRI images [9 - 11]. Fig 4 show the visual quality outcomes of M3IF-SSIM-CC on various datasets. Zooming into the images reveals that the results are good, and the new fusion rule applied based on similarity and correlation is effective. The edges and corners are effectively preserved. Uniformity is maintained, and no information is lost. It is evident that the M3IF-SSIM-CC is effective in fusing the CT and MRI image without loss of any information.

The results of M3IF-SSIM-CC are also compared with various standard methods as shown in Fig 4. The M3IF-SSIM-CC is compared with conventional methods like Principal Component Ananlysis (PCA), Discrete Wavelet Transform (DWT), Stationary Wavelet Transform (SWT), Wavelet Packet Decomposition (WPD), Multi-Singular Valued Dependency (MSVD), and NSCT, and the results are comparatively better than these traditional methods in terms of visual and qualitative analysis.



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(g) (h) (i)

Fig 4. (a) CT image (b) MRI image (c) MSVD (d) PCA (e) DWT (f) SWT (g) WPD (h) NSCT (i) M3IF-SSIM-CC

Upon thorough examination of the visual outcomes (Fig 4) and the parametric data (Figs 5, and 6), it can be concluded that the M3IF-SSIM-CC demonstrates superior performance relative to the other standard methods evaluated. The comparative performance of M3IF-SSIM-CC is better than all the conventional methods as shown in the Figs 4-6. It can be concluded that the application of this smart fusion rule with NSCT domain, the final fusion results are better. The fusion rule based on similarity and correlation is effective and can be improved outcomes if it may be use with the concept of method noise.

There is no reference image for the comparison, thus objective method performance evaluation measures are utilised to compare M3IF-NSST-MI to other techniques. The metrics employed include Functional Mutual Information (FMI), Fusion Symmetry (FS), Mean (M), Fusion Factor (FF), Standard Deviation (SD), Entropy (E), Fused Index (FI), and Sum of Correlation Difference (SCD). Except for FS, other measures should have greater values to get better outcomes. Only FS should have a lower value for optimal outcomes.

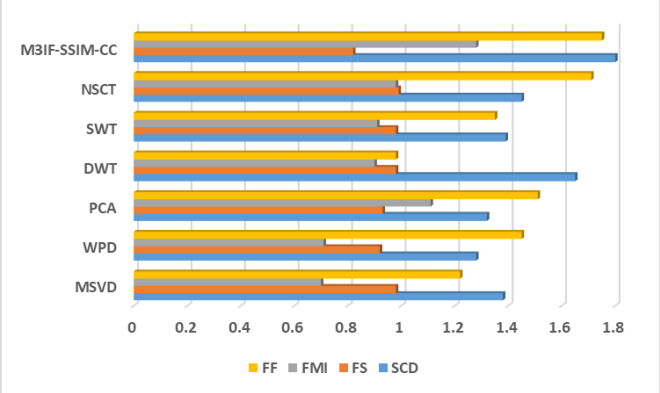


Fig 5. Graphical analysis based on the FF, FMI, FS, and SCD

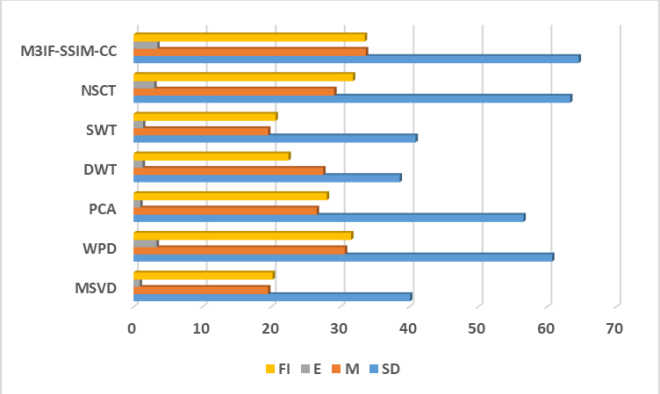


Fig 6. Graphical analysis of various metrics depicted in Table 1

Figures 5 and 6 provide graphical representations of the data of various parameters. This allows for fuller knowledge in terms of fine information preservation. The results shown in Figs 5 and 6, are averages of tests performed on over 75 datasets.

V. Conclusion and Future Scope

The M3IF-SSIM-CC method presents a robust approach for the integration of CT and MRI images, along with various other medical modalities, in a fusion process. This methodology enhances the quality of fused images through the application of the NSCT with SSIM and CC as used for fusion purpose. Following the execution of the NSCT on both CT and MRI images, the resulting wavelet coefficients are combined utilizing a sophisticated fusion rule to extract the most optimal information using SSIM and CC. The findings related to fusion are enhanced through the application of the smart fusion rule, which operates in conjunction with the maximal and average operation. This method enhances both the visual quality and parametric performance of the combined images. The M3IF-SSIM-CC method demonstrates significantly improved diagnostic utility and clarity in comparison to traditional fusion methods. This capability enhances clinical decision-making by generating fused images that exhibit greater accuracy and provide more informative insights.

Image fusion is only one of many possible uses for the wide variety of wavelet-based transformations that are in use today for processing and analyzing images. These transformations may be used to improve image analysis. Moreover, SSIM, CC, and more complex measures like Feature Similarity Indexing Method (FSIM) and Universal Image Quality Index (UIQI) may be used to modify and enhance the fusion rules. It is also possible to improve the fusion results even more by adding the idea of method noise as a post-processing procedure.

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